

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023- 2024 (Odd Semester)

Active Learning Practices

Degree, Semester & Branch: V Semester B.E.ECE A & B
Course Code & Title: CEC368 & IoT Based System Design
Name of the Faculty member: Mr.R.Ramasamy

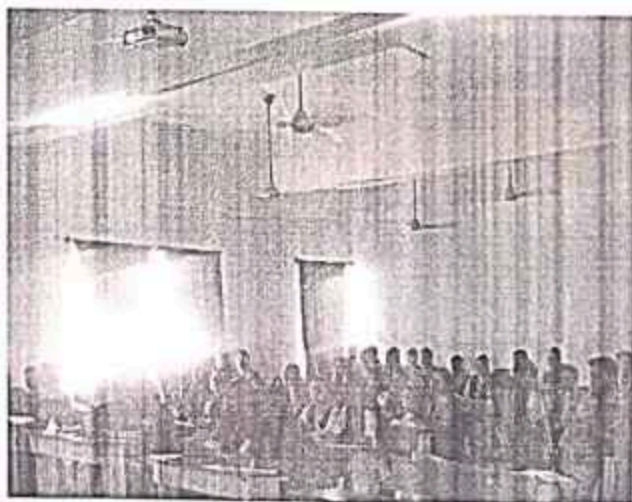
Active Learning Practices Execution

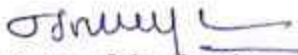
UNIT I INTRODUCTION TO INTERNET OF THINGS

Activity: One Minute Paper
Topic: IoT levels and deployment templates

- LEVEL: 1 953621106096
Shobin K.S
Single node
Local server
Analysis requirements are not computationally intensive.
- LEVEL: 2
Single node Eg: smart irrigation
Cloud server
Data stored in cloud app
Analysis requirements are not computationally intensive.
- LEVEL: 3
Single node.
Analysis requirements are computationally intensive.
Eg: Tracking and Package system
- LEVEL: 4
Multiple nodes
2- Observer nodes are used
Data size is big.

- IoT Level 1 Eg no: 953621106066
* Single node
* Analysis requirements are not needed.
- IoT Level 2
* Single node
* Analysis requirements are not needed.
- IoT Level 3
* Single node
* Analysis requirements are needed.
- IoT Level 4
* Multiple nodes
* Analysis requirements are needed.
- IoT Level 5
* Multiple nodes and one coordination node.




Signature of the faculty
05/08/2023


HOD.

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2022- 2023 (Even Semester)

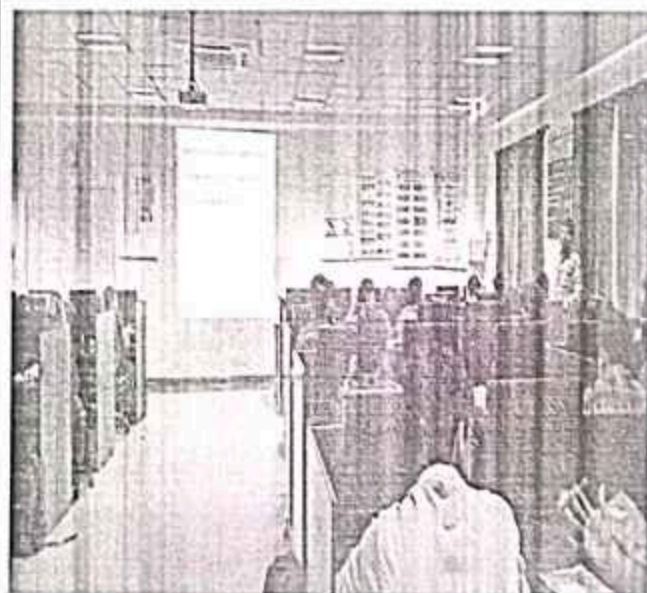
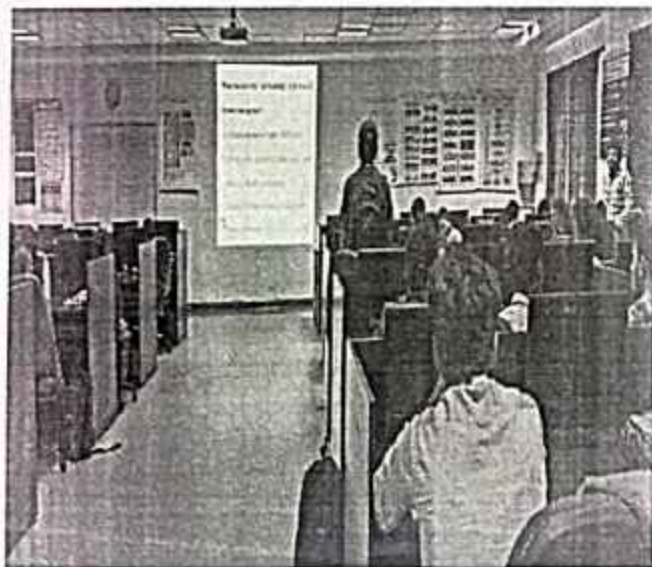
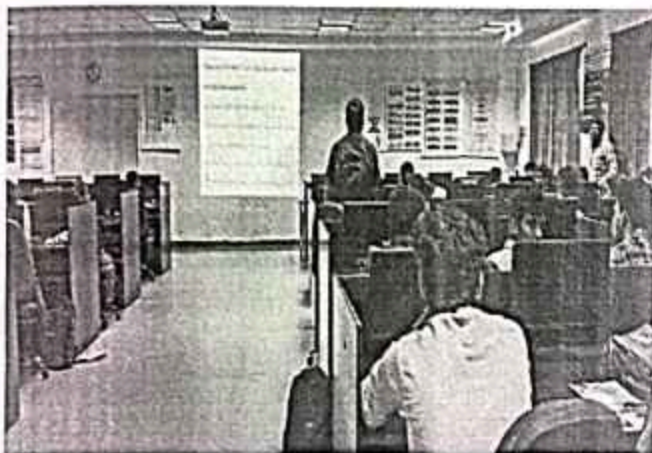
Active Learning Practices

Degree, Semester & Branch: V Semester B.E.ECE A & B
Course Code & Title: CEC368 & IoT Based System Design
Name of the Faculty member: Mr.R.Ramasamy

Active Learning Practices Execution

UNIT II MIDDLEWARE AND PROTOCOLS OF IOT

Activity: CLASS POLL
Topic: Resource Management in IoT



Signature of the faculty
04/09/2023

Signature
HOD

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2022- 2023 (Even Semester)

Active Learning Practices

Degree, Semester & Branch: V Semester B.E.ECE A & B
Course Code & Title: CEC368 & IoT Based System Design
Name of the Faculty member: Mr.R.Ramasamy

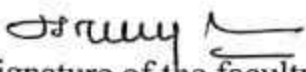
Active Learning Practices Execution

UNIT III COMMUNICATION AND NETWORKING

Activity: Role Play

Topic: Application Layer Protocols: CoAP and MQTT, Data aggregation & dissemination.




Signature of the faculty
07/10/23


HOD

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023- 2024 (Odd Semester)

Active Learning Practices

Degree, Semester& Branch: V Semester B.E.ECE A & B
Course Code & Title: CEC368 & IoT Based System Design
Name of the Faculty member: Mr.R.Ramasamy

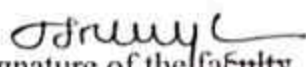
Active Learning Practices Execution

UNIT IV IOT IMPLEMENTATION TOOLS

Activity: Jigsaw

Topic: Implementation of IoT with Raspberry Pi




Signature of the faculty
06/10/23


HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2023-2024 (Odd Semester)

Planning Document

Degree, Semester & Branch: V Semester B.E. ECE A & B

Course Code & Title: CEC368 & IoT Based System Design

Name of the Faculty member: Mr.R.Ramasamy, AP/ECE

- **Unit/Topic:** Unit-IV / Implementation of IoT with Raspberry Pi
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 14
- **Activity Chosen:** Jigsaw

Learning Outcomes:

The Student will be able to

- ❖ Understand the concept of Implementation of IoT with Raspberry Pi
- ❖ Know the concept of Raspberry Pi Interfaces
- ❖ Participate actively in the learning process
- ❖ Discuss the concepts, convey idea and share the views of Raspberry Pi GPIO, Raspberry Pi Interfaces, Interfacing LED and switch with Raspberry Pi, Interfacing a Light Sensor (LDR) and switch with Raspberry Pi
- ❖ Communicate effectively by sharing their views among the teammates

Justification:

The combination of IoT with Raspberry Pi offers a cost-effective, versatile, and user-friendly platform that is well-supported by a vibrant community. These factors make it a compelling choice for a wide range of IoT applications, from hobbyist projects to industrial solutions. The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. The activity 'JIGSAW' is used to recall the concepts learnt and enhances the learning through this collaboration

- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**

(i) Materials for the Activity:

The materials for the preparation of the students will be shared one week before through LMS Canvas. Including the material, students were asked to refer to the web content also.

References:

Book Title/ Author/ Publisher/ Edition	Page No.
1. Vijay Madisetti and Arshdeep Bahga, "Internet of Things (A Hands-onApproach)", VPT, 1st Edition, 2014.	179-195
Websites	
1. https://www.youtube.com/watch?v=mj9JIqNjj6E	
2. https://www.youtube.com/watch?v=VZcKLl2hokk	
3. https://www.youtube.com/watch?v=N9rg5nKrKQU&t=697s	

(ii) Formation of Groups:

2. How do you rate the reference material provided related to the topic? Give tick mark (✓)

Excellent	Good	Satisfactory	Poor
✓			

3. Have you enjoyed the activity? Give tick mark (✓)

Yes: ✓

No:

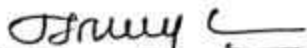
4. Challenges faced during activity/presentation. Give tick mark (✓)

Yes: ✓

No:

If yes what?:

B. Rithika 25/10/23
Sundar M 29/10/23
B. RITHIKA, K.S. Muthu Sathya
Muthu Sathya 26/10/23
Name & Signature of the Student with date


Signature of Faculty Member
26/10/2023


HOD

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023- 2024 (Odd Semester)

Active Learning Practices

Degree, Semester & Branch: V Semester B.E.ECE A & B
Course Code & Title: EEC368 & IoT Based System Design
Name of the Faculty member: Mr.R.Ramasamy

Active Learning Practices Execution

UNIT V APPLICATIONS AND CASE STUDIES

Activity: One Minute Paper
Topic: Industry applications of IoT

Industry Application
Industry application of IoT:

Open and use applications in IoT industry

1. Monitor temperature & pressure
2. Reduce data quality transmission

Monitor temperature and pressure:

1. To determine the cause of machine fault.
2. Monitor and the machine can monitor the operating conditions such as (temperature and pressure levels)

Reduce data quality transmission:

1. Reducing data in quality features is expected both in data and data quality.
2. Monitor data quality such as
 - a. Low - constant transmission
 - b. Low - irregular transmission
 - c. High - irregular transmission
 - d. High - constant transmission

Monitor data quality.



Signature of the faculty
 07/11/2023

HOD